



TECHNICAL ARCHITECTURE PAPER

Composable ERC Standards for Titled Asset Tokenisation

A Composable Standard Architecture for
Titled Asset Tokenisation on EVM

Composable ERC Standards for Titled Asset Tokenisation

A Composable Standard Architecture for Titled Asset Tokenisation on EVM

Six ERC proposals addressing the structural gap between existing token standards and the requirements of institutional real-world asset issuance:

1. Asset-to-token binding
2. Directional transfer domains
3. Deterministic document anchoring
4. On-chain impact attestation
5. Standardised Net Asset Value (NAV) reporting
6. On-chain compliance event recording

Field	Value
Author	Kula
Date	May 2026
Status	Non-normative companion documentation
Classification	Public pre-submission draft
Academic Foundation	Turner, C. (2026). Where Authority Resides: A Completion-Based Taxonomy of Tokenisation. SSRN.

Contents

Composable ERC Standards for Titled Asset Tokenisation	2
Contents	3
1. Introduction	5
1.1 The Emerging Jurisdictional Landscape	6
1.2 What This Paper Is	7
2. Theoretical Foundation	8
3. The Regulatory Imperative	10
3.1 SFDR and Product-Level Sustainability Disclosure	10
3.2 EU Taxonomy Regulation and Classification Standards	10
3.3 ISSB IFRS S2 and Climate-Related Financial Disclosures	11
3.4 Compliance Event Infrastructure: FATF Travel Rule and Regulatory Reporting	11
3.5 Jurisdictional DLT Securities Frameworks	12
4. The Existing Standards Landscape	13
4.1 Fungible Token Infrastructure	13
4.2 Compliance and Permissioned Transfer	13
4.3 Data and Identity	15
4.4 Title and Asset Representation	15
4.5 Non-EVM and Cross-Platform Production Deployments	16
4.6 The Composite Gap	18
5. Cross-Ecosystem Comparison: EVM and XRPL	22
6. The Architectural Response: Six Composable ERC Proposals	25
6.1 Proposal 1: Asset Anchor Registry Interface	27
6.2 Proposal 2: Canonical Document Bundle Anchor	29
6.3 Proposal 3: Directional Transfer Domain Registry	31
6.4 Proposal 4: Subject Linked Compliance Event Log	33
6.5 Proposal 5: Subject Linked Impact Snapshot Log	35
6.6 Proposal 6: Subject Linked NAV Snapshot Oracle	37
6.7 Composition	39

6.8 Minimal Adoption Path	39
7. Deployment Economics	41
8. The Case for Standards Over Bespoke Infrastructure	43
8.1 Risks to Adoption	43
9. Conclusion	45
Annex 1 - Why these six standards?	49
How the Six Standards Compose	49
Asset Anchor Registry Interface	52
Canonical Document Bundle Anchor	54
Directional Transfer Domain Registry	56
Subject-Linked Compliance Event Log	58
Subject-Linked Impact Snapshot Log	60
Subject-Linked NAV Snapshot Oracle	62

1. Introduction

Tokenised real-world assets on public blockchains are approaching US\$31 Billion in on-chain value (RWA.xyz, 2026). The trajectory is clear: BlackRock, Franklin Templeton, Backed Finance, Ondo, and Centrifuge are issuing tokenised treasury bills, money-market fund shares, and private credit instruments on Ethereum, Avalanche, and Polygon. Institutional capital is arriving.

BridgeTower Capital's \$11 billion DOM X tokenisation on Avalanche illustrates the current state of the potential of tokenisation with a successful, vertically integrated deployment built on proprietary Chainlink components. It works. What it does not provide is open, interoperable interfaces.

The current Ethereum token standards landscape provides the building blocks for compliant token issuance. ERC-20 defines fungible token mechanics. ERC-3643 adds identity-gated transfer compliance. ERC-7208 introduces composable on-chain data objects. ERC-7943 reached final status in May 2026 and standardises a minimal vendor-neutral compliance interface. These are real, functional, widely adopted, or early in the pipeline standards.

None of them, individually or composed together, addresses the structural problem that institutional RWA issuance requires a solution for, namely binding a compliant fractional instrument to a specific, identifiable, legally anchored underlying asset in a way that is enforced by the contract rather than asserted by the issuer.

This white paper presents the architectural rationale for a family of six ERC proposals developed by Kula. Each proposal addresses a specific gap that no published ERC standard surveyed provides. Each stands independently and is designed to compose with the others. None replaces existing standards. Each extends them into territory they were not designed to cover.

The core primitive in the family is Proposal 1, which defines a minimal registry interface for binding token contracts to claims about off-chain real-world assets. The standard separates the problem into two interfaces. The registry interface (IAssetAnchorRegistry) records anchor entries with dual hash fields (legalHash for the legal basis, evidenceHash for the custody or evidence claim), enforces one-to-one binding within the registry domain, and prevents rebinding after a token is anchored. The token interface (IAssetBoundContract) allows any token contract to declare its binding to a registry anchor and expose it through a standard interface call. This makes asset-to-token binding structurally verifiable rather than merely asserted in metadata. The standards survey in Section 4 identifies no existing ERC standard that provides this exact combination of structural separation, mutual binding, and registry-enforced exclusivity. The remaining five Proposals address document verification (Proposal 2), directional transfer domains

(Proposal 3), compliance event recording (Proposal 4), impact attestation (Proposal 5), and valuation reporting (Proposal 6).

1.1 The Emerging Jurisdictional Landscape

Multiple jurisdictions are now deploying live infrastructure for tokenised titled assets, each using bespoke stacks because no common standard exists.

In Dubai, the Dubai Land Department (DLD) tokenised property title deeds on the XRP Ledger via PRYPCO Mint, in coordination with Ctrl Alt, synchronised with the official land registry. Secondary market trading launched in early 2026 within a regulated pilot framework.

In Switzerland, the DLT Act (2021) created ledger-based securities (Registerwertrechte) as a legally recognised instrument class. BX Digital, SIX Digital Exchange, AMINA Bank, 21X, and Tokeny have together built a complete regulated pathway from traditional securities to on-chain trading using ERC-3643 as the issuance standard.

In Germany, the Electronic Securities Act (eWpG, 2021) allows electronic securities to be recorded on crypto securities registers supervised by BaFin. ERC-7551 was developed specifically for eWpG compliance, covering bonds and fund shares. Siemens issued a €300 million corporate bond on-chain under the framework.

Each of these jurisdictions has deployed live infrastructure for tokenised titled assets using bespoke stacks with no common standard and no interoperability between them.

Thesis

No composition of existing ERC standards produces the structural binding, directional transfer-domain rules, deterministic document verification, auditable impact attestation, standardised valuation reporting, and compliance event recording that institutional titled-asset issuance requires. The six proposals address these gaps. Each stand independently and composes with the others.

1.2 What This Paper Is

This is the non-normative companion document referenced by the six Proposal specifications. It explains the motivation, theoretical foundation, and architectural rationale for the proposals. It maps the existing standards landscape exhaustively. It positions the six Proposals against comparable capabilities on other ledgers. It does not repeat the formal specifications, which are contained in the Proposals themselves.

The paper is structured as follows.

Section 2 establishes the theoretical foundation from the completion authority framework found in “Where Authority Resides” (2026) that identifies where in the progression from referential to atomic tokenisation the current gap sits.

Section 3 maps the regulatory drivers making this gap urgent.

Section 4 surveys the existing ERC standards landscape and identifies precisely where each standard stops.

Section 5 provides a cross-ecosystem comparison with XRPL’s native capabilities.

Section 6 presents the six-Proposal architecture.

Section 7 analyses deployment economics.

Section 8 builds the case for standards over bespoke solutions.

Section 9 concludes.

2. Theoretical Foundation

The analytical framework for this architecture is drawn from the tokenization taxonomy “Where Authority Resides” (2026), which proposes completion authority as the single organising variable for a taxonomy of tokenisation. The framework identifies four categories, distinguished by where the authority to complete a transaction resides relative to the ledger record.

Table 1: Theoretical Taxonomy

Category	Completion Authority	Token's Relationship to Asset	Examples
Referential	Entirely external. Token is an identifier, not a claim.	Points to the asset. Conveys no rights.	Supply chain tracking tokens, provenance markers, non-financial NFTs
Contractual	External, mediated by legal agreement. Token represents a contractual entitlement.	Represents a claim enforceable under contract law. The token holder has rights, but those rights are completed by an intermediary.	ERC-3643 security tokens, CMTAT instruments, tokenised fund shares, stablecoins
Title	Partially internal. The ledger is legally recognised as a record of title, but institutional frameworks still govern the recognition.	Is the record of title. Transfer on the ledger constitutes transfer of the legal interest, within the scope of the recognising framework.	Swiss Registerwertrechte, German eWpG electronic securities, Dubai Land Department real estate deed tokens (DLD/PRYPCO Mint/ Ctrl Alt, XRPL)
Atomic	Fully internal. The ledger is the sole source of truth. No external institution is needed to complete the transaction.	Is the asset. Transfer is final, irreversible, and self-executing.	Native cryptocurrency (BTC, ETH). No RWA has achieved this status. Ref. Network State.

The taxonomy reveals a structural observation. Existing ERC standards have followed an unintentional but coherent progression toward internalising completion authority. ERC-20 internalised accounting. ERC-721 internalised identification. ERC-3643, ERC-7518, and ERC-7943, each internalised validation, in their own way - ERC-3643 through ONCHAINID and attached compliance modules, ERC-7518 through partition-based rules and dynamic compliance vouchers, and ERC-7943, by distilling compliance to a minimal vendor-neutral interface of transfer checks and enforcement actions. However, none of these standards internalizes recognition, where the ledger's record is treated as legally authoritative with respect to an identified real-world asset. ERC-3643 is explicit about this. It defines how a compliant token behaves, not what it represents. ERC-7943, makes the same boundary visible from the other direction, since its forced-transfer primitive exists precisely to execute an external legal mandate the contract does not itself constitute. In every case the token's relationship to an underlying asset is defined outside the contract, in legal agreements the contract does not reference, verify, or enforce.

For native digital assets such as stablecoins, governance tokens, utility tokens this standard is sufficient. The token is the asset, or the token's relationship to its backing is simple enough that a single contractual framework covers it.

For titled real-world assets such as real estate, infrastructure concessions, natural resource rights, structured credit instruments, the standard is not sufficient. These assets have legal title that exists independently of any ledger. Fractionalising them requires maintaining the legal unity of title while issuing divisible economic claims against it. Transferring those claims across jurisdictions requires directional corridor rules that individual KYC checks cannot provide. Verifying the legal documents that anchor the on-chain representation requires deterministic hashing that current standards do not specify. Measuring the impact those assets generate requires structured, auditable, append-only data infrastructure that does not exist on-chain (at the institutional level impact reporting requirements are a critical component in asset management). Reporting their valuation requires a standardised on-chain NAV interface that no ERC defines. And recording the compliance events that accompany every regulated action requires a structured, queryable event schema that the entire ecosystem currently implements in isolation.

The six ERC proposals presented in this paper occupy the space between contractual and title tokenisation in the referenced taxonomy. They do not claim to achieve title tokenisation, which requires statutory recognition and is a legislative achievement, not a technical one. They specify the on-chain architecture that jurisdictions extending legal recognition to ledger-based records of title would need to be in place.

Scope Boundary

This architecture operates at the boundary between contractual and title tokenisation. It specifies the technical substrate with binding, compliance, document verification, impact attestation that statutory title recognition frameworks require. It does not and cannot substitute for the legal frameworks themselves. Implementers **MUST** obtain independent legal advice for each jurisdiction in which instruments are issued or transferred.

3. The Regulatory Imperative

Five regulatory developments are converging to make the absence of on-chain compliance, impact, and transfer metadata infrastructure an urgent commercial problem, not merely a theoretical gap. Institutional adoption of the tokenisation of real-world assets will be predicated on substantive solutions to the following regulatory requirements.

3.1 SFDR and Product-Level Sustainability Disclosure

The EU's Sustainable Finance Disclosure Regulation (SFDR) requires financial market participants to disclose sustainability information at both entity and product level. SFDR Level 2 requires product-level Principal Adverse Impact (PAI) indicator reporting for Article 8 and Article 9 financial products. The proposed SFDR 2.0 revision strengthens product-level obligations and moves toward more granular, asset-level sustainability data.

For tokenised RWA products marketed in the EU, or to EU-domiciled investors, this means that the underlying assets must produce structured, auditable sustainability data that can be aggregated at the product level. Currently, this data lives in PDF reports produced by the issuer. There is no standard mechanism for independent verification, no structured format for cross-product comparison, and no technical barrier to retroactive modification.

3.2 EU Taxonomy Regulation and Classification Standards

The EU Taxonomy Regulation (2020/852) establishes the classification framework that determines what qualifies as "environmentally sustainable" for the purposes of SFDR disclosure. Financial products marketed as Article 8 or Article 9 must report their Taxonomy alignment, which requires asset-level assessment against six environmental objectives and the "do no significant harm" criteria.

For tokenised assets backed by physical infrastructure, the Taxonomy assessment must be traceable to the specific asset and its measurable environmental performance, not to a portfolio-level estimate. A tokenised solar installation must demonstrate its contribution to climate change

mitigation through verifiable energy generation data. A tokenised forestry concession must demonstrate biodiversity protection through measurable habitat outcomes. The Taxonomy does not prescribe how this data should be recorded or verified, but it assumes that the data exists, is auditable, and can be independently assessed. Proposal 5's structured impact attestation layer is designed to meet exactly this requirement.

3.3 ISSB IFRS S2 and Climate-Related Financial Disclosures

The International Sustainability Standards Board's IFRS S2 requires entities to disclose information about climate-related risks and opportunities. Jurisdictions adopting ISSB standards, including the UK, Australia, Singapore, and others, will require Scope 1, 2, and 3 emissions reporting with increasing granularity. For tokenised assets backed by physical infrastructure (energy generation, agriculture, real estate), the emissions data must be traceable to the specific asset, not aggregated at the portfolio level.

3.4 Compliance Event Infrastructure: FATF Travel Rule and Regulatory Reporting

The Financial Action Task Force's Recommendation 16, commonly known as the Travel Rule, requires Virtual Asset Service Providers to exchange originator and beneficiary information when transferring virtual assets above a de minimis threshold. The rule has been implemented, or is being implemented, across over 50 jurisdictions worldwide.

In the EU, the Travel Rule is given legislative force through the Transfer of Funds Regulation (TFR, Regulation 2023/1113), adopted as a companion measure to the Markets in Crypto-Assets Regulation (MiCA). The TFR requires that originator and beneficiary information accompanies every crypto-asset transfer, with no de minimis threshold for transfers between VASPs. This is stricter than the FATF baseline and applies to all crypto-asset service providers licensed under MiCA.

The regulatory pressure is immediate and universal. Singapore's Payment Services Act, Japan's Payment Services Act, South Korea's Virtual Asset Users Protection Act, and comparable frameworks across the G20 all mandate Travel Rule compliance. For institutional RWA tokenisation, where every transfer crosses jurisdictional boundaries and involves regulated counterparties, Travel Rule compliance is not an optional feature. It is a licensing prerequisite.

There is no ERC standard for expressing structured compliance event metadata on-chain. Every platform implements compliance event recording through proprietary off-chain systems. The result is that the on-chain transfer record and the compliance metadata that regulators require exist on separate rails with no standard mechanism for cross-referencing them. Proposal 4

addresses this directly, with Travel Rule-related records representable through transfer payloads or custom event and payload profiles within the broader compliance event schema.

3.5 Jurisdictional DLT Securities Frameworks

Two jurisdictions have enacted frameworks that explicitly recognise distributed ledger records as legally authoritative for certain classes of securities:

- Switzerland: the Federal Act on the Adaptation of Federal Law to Developments in Distributed Ledger Technology (DLT Act, 2021) recognises ledger-based securities (Registerwertrechte) as a distinct category. The Capital Markets and Technology Association (CMTA) has published the CMTAT standard for Swiss-law tokenised securities.
- Germany: the Electronic Securities Act (eWpG, 2021) allows electronic securities to be recorded on a crypto securities register. ERC-7551 was developed specifically for eWpG compliance. BaFin supervises crypto securities registrars.

The EU's MiCA framework, while not a securities law, establishes the broader regulatory environment within which tokenised assets operate in Europe. MiCA creates licensing requirements for crypto-asset service providers, mandates white paper disclosures for asset-referenced tokens, and requires issuers of asset-referenced tokens to maintain and disclose reserve asset valuations. This last requirement directly supports the case for a standardised on-chain NAV interface (Proposal 6).

Both DLT securities frameworks sit in the title tokenisation category of the tokenisation taxonomy introduced above. The ledger is legally recognised as authoritative, but only within the scope of the recognising statute. Both still depend on external institutional frameworks for their validity. The boundary the taxonomy identifies is institutional, not technical, and holds in both cases.

4. The Existing Standards Landscape

This section maps every ERC standard and comparable framework relevant to real-world asset tokenisation, identifies what each solves, and locates precisely where it stops. The analysis is structured by the problem each standard addresses rather than chronologically, because the gaps become visible only when the problems are seen as a connected system.

4.1 Fungible Token Infrastructure

ERC-20

The foundational fungible token standard. Defines balances, transfers, approvals, and allowances. Every subsequent token standard either extends or composes with ERC-20. It has no concept of compliance, identity, asset binding, or governance. It is plumbing, not architecture. Everything in this paper builds on it.

ERC-4626: Tokenised Vaults

A standard for yield-bearing vaults. Assets go in, fractional share tokens come out. This is the closest thing in DeFi to the wrapper concept that institutional RWA tokenisation requires. But it is entirely DeFi-native with no legal anchoring, no compliance layer, no title reference, and no jurisdiction field. It solves the economic structure without the legal architecture.

4.2 Compliance and Permissioned Transfer

ERC-1400 / ERC-1404

Early security token standards. ERC-1400 introduced partitioned balances and document management. ERC-1404 added transfer restriction detection. Both contributed conceptually but neither achieved broad adoption. ERC-3643 superseded them as the institutional compliance standard.

ERC-3643 (T-REX)

The most mature compliance token standard. ERC-3643 embeds identity verification via ONCHAINID, transfer restrictions, investor eligibility rules, and agent-based forced transfer directly into the token contract. It is the de facto standard for compliant security tokens and has been adopted by Tokeny, Fireblocks, and others.

What ERC-3643 solves is compliant token issuance with identity-gated transfers, freeze, forced transfer, and agent management. What it does not solve is the token's relationship to an underlying asset. ERC-3643 defines how a compliant token behaves. It does not specify what

the token represents, how its economic claim is bound to a specific titled asset, or how that binding is enforced. The compliance is at the instrument layer where the asset layer is outside its scope.

This is the critical boundary. ERC-3643 is a contractual tokenisation standard in the tokenisation taxonomy introduced above. The token represents a claim, and the claim's validity depends on legal agreements that exist outside the contract. The six Proposals in this paper do not replace ERC-3643. They extend it into the asset layer that it deliberately leaves unspecified.

ERC-7943: uRWA (Universal Real-World Asset Interface). ERC-7943 (uRWA) reached Final status in May 2026 with a coalition of supporters including Brickken, Hacken, CMTA, DigiShares, Stobox, and QuillAudits, and has been integrated into CMTAT and demonstrated compatible with Chainlink ACE. This standard is a minimal universal compliance interface for RWA tokens. Defines five functions: `forcedTransfer`, `setFrozenTokens`, `getFrozenTokens`, `isUserAllowed`, and `canTransfer`. It is designed to be maximally compatible across ERC-20, ERC-721, ERC-1155, and ERC-6909.

ERC-7943 is deliberately unopinionated. It standardises the external compliance interface without prescribing identity frameworks, metadata structures, or enforcement mechanisms — leaving those choices to implementers and issuers.

ERC-7943 serves as the external compliance interface on a token. Separate asset-binding standards like ERC-7929 (PermaLink) address token-to-token hierarchies, a different problem domain.

ERC-7551

Created specifically for compliance with Germany's eWpG framework. Defines interfaces for electronic securities with regulatory-specific features including securities registration, regulated agent roles, and German-law transfer semantics. Narrower jurisdictional scope than the architecture presented here, but important as evidence that jurisdiction-specific token standards are emerging and will need to interoperate.

CMTAT

The Capital Markets and Technology Association Token standard. An open framework for smart contracts designed for tokenisation of financial instruments, maintained by a Swiss not-for-profit with members from Taurus, Fireblocks, Bitcoin Suisse, Swissquote, and Sygnum. CMTAT implements ERC-20, integrates with ERC-3643, ERC-7943, ERC-7929, ERC-1404, and ERC-7551, and includes cross-chain functionality via ERC-7802. It has been audited twice and is referenced in MAS Project Guardian.

CMTAT covers compliant security token issuance, conditional transfers, freeze, forced transfer, snapshot-based distributions, upgradeability via proxy, and corporate actions. What it does not cover: title anchor binding (no titleHash, no registryId, no legalCustodian field), the wrapper architecture (no holding vehicle concept), structured clawback with trigger types and legal reference hashes, or impact data of any kind. CMTAT represents the instrument. This architecture represents the relationship between the instrument and the thing it represents.

4.3 Data and Identity

ERC-7208: On-Chain Data Objects

Introduces composable on-chain data as a first-class concept. Data Objects hold state, Data Managers define business logic, Data Points are key-value pairs with a canonical addressing scheme. This is the most architecturally significant standard for this paper's purposes because it provides the data layer on which asset-level metadata, impact data, and document references can be structured.

What ERC-7208 provides is a composable, upgradeable data layer where ownership of data is separated from the logic that operates on it. What it does not specify is how title to a physical or legal asset should be structured, what canonical keys an asset record should contain, or how fractional instruments should relate to asset data. ERC-7208 is infrastructure where this architecture is one application of that infrastructure.

ERC-735 / ONCHAINID / EAS

Identity and attestation standards. ERC-735 proposed on-chain claims in 2017 and stagnated. ONCHAINID, developed by Tokeny for ERC-3643, is the most widely deployed on-chain identity system for compliant tokens. The Ethereum Attestation Service (EAS) provides a generic attestation framework widely adopted since 2023.

The identity gap relevant to this architecture: ERC-3643's coupling to ONCHAINID limits interoperability with other identity systems. W3C Decentralised Identifiers (DIDs) are now a formal standard with multiple EVM implementations (did:ethr, did:pkh, did:key). XRPL has its own DID implementation. An architecture designed for multi-jurisdictional RWA issuance needs identity agnosticism, or the ability to work with any compliant identity backend without requiring a specific registry. The Kula reference implementation addresses this through an IdentityResolver interface pattern, which is compatible with ONCHAINID, did:ethr, Polygon ID, and other W3C DID methods.

4.4 Title and Asset Representation

ERC-721: Non-Fungible Tokens

Represents unique asset ownership. Widely used for digital collectibles but also explored for title representation. The fundamental limitation for RWA use: ERC-721 tokens cannot be fractionalised without destroying the legal unity of title. One token, one owner. There is no mechanism to issue divisible economic claims against an ERC-721 while maintaining the indivisibility of the underlying title.

ERC-6065: Real Estate Title Tokenisation

Extends ERC-721 with a hash of the operating agreement, unique property identifiers, debt value, foreclosure status, and a manager address. The closest existing standard to title tokenisation. But it inherits ERC-721's limitation: one token, one owner, no fractionalisation mechanism, no compliance layer, no wrapper concept.

ERC-6956: Asset-Bound NFTs

Defines NFTs bound to physical assets via anchor certificates. Closer to the binding concept in this architecture but limited to non-fungible tokens. No fractionalisation. No compliance. The binding is one-to-one, not one-to-many.

ERC-7929: PermaLink Asset Bound Token

Defines PermaLink-ABTs permanently bound to a parent token via a one-way `assetBoundContract` link, with optional reveal-from-supply mechanics. Supports nested/portfolio-like RWA structures where subordinate assets move with a binding token. Distinction from Proposal 1: hierarchical parent↔child token binding, not a standalone registry with dual-hash off-chain anchors, mutual registry↔token verification, or fungible whole-contract binding. Complementary for nested structures; Proposal 1 addresses registry-mediated anchoring for primary asset identity.

ERC-1155 / ERC-7518 (DyCIST)

ERC-1155 is a multi-token standard. ERC-7518 extends it with partitions where each token ID represents a unique asset type or investor group, each with its own rights, restrictions, and metadata. Partitions approach the cell concept: multiple asset classes within a single contract. But partitions are asset class differentiators, not legally distinct holding vehicles for separate title references. The wrapper insight, converting a property problem (indivisible title) into a contract problem (holding vehicle issues divisible claims), is not present.

4.5 Non-EVM and Cross-Platform Production Deployments

The ERC landscape does not exist in isolation. Titled asset tokenisation is already in production on multiple ledgers and across multiple infrastructure configurations. Two categories of deployment are particularly instructive because they demonstrate both the operational maturity of the field and the architectural questions that emerge when production systems operate without a common standard.

Government-backed real estate title tokenisation (XRPL)

The most advanced production deployment of titled real estate tokenisation globally is a government land registry integration in the Middle East, where fractional deed tokens on the XRP Ledger are synchronised in real time with the official property registry. Phase One tokenised ten properties representing over \$5 million in value. Phase Two introduced secondary market trading within a regulated sandbox. The issuing authority projects \$16 billion in tokenised real estate by 2033.

The deployment addresses several of the architectural requirements identified in this paper directly. Asset-to-title binding is achieved through live synchronisation between on-chain tokens and the government registry. Compliance is managed through a structured regulatory framework involving the virtual asset regulator, a government innovation authority, and a dedicated real estate sandbox. Fractional ownership is operational. Secondary trading is in pilot. The XRPL's native protocol features, including clawback, permissioned domains, and decentralised identifiers, provide infrastructure at the ledger layer that EVM currently lacks as application-level standards.

What the deployment does not address is portability beyond its own ecosystem. The architecture is purpose-built for a single registry and a single ledger's native features. A second jurisdiction seeking to replicate the model would need to build equivalent registry integration, compliance infrastructure, and secondary market mechanics independently. The question is not whether the deployment works. It clearly does. The question is whether the architectural decisions it embodies are capturable in a standard that others can adopt.

Large-scale natural resource tokenisation (EVM / oracle infrastructure)

An \$11 billion natural resource asset, a copper-gold project in the United States, has been tokenised on an EVM-compatible chain using a vertically integrated oracle and infrastructure stack. The deployment uses cross-chain interoperability protocols for multi-chain settlement, proof-of-reserve verification for asset backing, an on-chain NAV feed for real-time valuation, and a runtime environment for compliance and settlement automation. This is live production infrastructure, not a pilot.

The deployment addresses asset verification, on-chain valuation, cross-chain settlement, and compliance automation. The infrastructure is production-grade, audited, and institutionally credible. The issuer has indicated a pipeline exceeding \$25 billion in additional natural resource, energy, and metals assets.

The architectural observation parallels the real estate deployment. The solution is vertically integrated around a single infrastructure provider's stack. A project seeking comparable functionality without adopting the same stack would need to source or build equivalent components independently. This is not a deficiency. Vertical integration delivers speed, coherence, and accountability. It is an observation about what happens at ecosystem scale when multiple issuers, jurisdictions, and infrastructure providers need to interoperate. Vertical stacks compose within themselves. Standards compose across the ecosystem.

The common observation

Both deployments confirm that the capability to tokenise titled real-world assets at institutional scale exists today, on multiple ledgers, with production-grade infrastructure. Both also illustrate that each deployment is architecturally self-contained. The real estate deployment does not interoperate with the natural resource deployment. Neither interoperates with the regulated European DLT trading and settlement systems now operating under Swiss and EU frameworks. Each is a complete, functional, isolated vertical. The six ERC proposals in Section 6 do not claim to improve on what any of these deployments have built. They propose the common interface layer that would allow future deployments to compose with each other rather than operating in parallel.

4.6 The Composite Gap

The following table maps the seven architectural requirements of institutional titled-asset tokenisation against every standard surveyed. No standard, and no composition of standards, covers all six.

Several further standards sit adjacent to this family and are referenced in the prior-art sections of the individual specifications. For transfer compatibility and restriction, ERC-1404, ERC-1462, and ERC-1592 define token-local transfer-rule and restriction interfaces, and ERC-7802 addresses cross-chain token bridging; Proposal 3 operates as a token-agnostic registry above these. For document handling, ERC-1643 defines document management for security tokens and ERC-5289 defines a notary interface; Proposal 2 differs in providing deterministic canonical bundle hashing rather than reference storage. For valuation, ERC-7726 defines a common quote oracle for liquid assets, while ERC-7540 and ERC-7575 extend ERC-4626 for asynchronous and multi-asset vaults; Proposal 6 addresses periodic illiquid NAV that these do not. For compliance, ERC-

8106 proposes an event-based compliance framework using soft advisory events and entity classification, where Proposal 4 instead provides structured append-only records with authority attribution and evidence hashes. For identification, ERC-8127 defines human-readable token identifiers adjacent to Proposal 1's asset anchors. ERC-7943 has additionally been submitted to the SEC Crypto Task Force as part of its author's regulatory engagement, indicating the policy relevance of standardised compliance primitives.

Table 2: Prior Art and Proposal Gaps

Standard	Asset Binding	Document Verification	Directional transfer-domain rules	Compliance Events	Impact Attestation	NAV Reporting
ERC-20	X	X	X	X	X	X
ERC-721	Partial (unique ID)	X	X	X	X	X
ERC-3643	X	X	Unilateral (KYC only)	X	X	X
ERC-7208	Data layer only	X	X	X	X	X
ERC-7943	X	X	Unilateral (checks only)	X	X	X
ERC-6065	Title hash, no fractions	X	X	X	X	X
ERC-6956	Anchor cert, NFT only	X	X	X	X	X
ERC-7518	Partitioned, not title-bound	X	X	X	X	X
ERC-7551	German eWpG only	X	German law only	X	X	X
CMTAT	X	X	Via ERC-3643	X	X	X
ERC-4626	Vault wrapper, no legal anchor	X	X	X	X	Bespoke per vault
This Architecture	✓ Proposal 1	✓ Proposal 2	✓ Proposal 3	✓ Proposal 4	✓ Proposal 5	✓ Proposal 6

No standard in the table, and no composition of them, addresses all six dimensions simultaneously.

BridgeTower Capital's DOM X deployment, described in the Introduction, is the clearest illustration of what bespoke build looks like at institutional scale. Four proprietary Chainlink components assembled into a custom stack because no standard interface existed for any of the required functions. The six Proposals presented in Section 6 address each of those gaps as open, composable, vendor-neutral interfaces.

5. Cross-Ecosystem Comparison: EVM and XRPL

The architectural gap identified in Section 4 is specific to the EVM ecosystem. XRPL, the XRP Ledger, addresses several of these requirements at the protocol layer, making a cross-ecosystem comparison both technically instructive and strategically relevant.

This is not a competitive analysis in the conventional sense. XRPL and EVM are architecturally different systems with different design philosophies. XRPL builds compliance, identity, and transfer control into the protocol itself. EVM provides a general-purpose computation layer and expects application-level standards to handle compliance. Neither approach is inherently superior; they make different trade-offs between protocol-level enforcement and application-level flexibility.

The comparison matters because institutional RWA issuers will deploy on the infrastructure that meets their requirements. Ethereum currently hosts over 56% of all distributed tokenised asset value globally, with approximately \$17 billion on mainnet alone. When EVM-compatible chains are included (Avalanche, Polygon, Arbitrum, Base, Optimism), the EVM ecosystem accounts for the overwhelming majority of institutional RWA deployment. XRPL's tokenised asset base has grown rapidly but remains substantially smaller in distributed value. The power centre of institutional tokenisation is on EVM. The risk is not that EVM loses that position to XRPL in aggregate, but that specific high-value use cases with demanding compliance requirements, such as titled real estate and regulated natural resource instruments, migrate to XRPL because it provides the compliance primitives natively while EVM requires them to be assembled in a bespoke manner. Standardising those primitives on EVM preserves the ecosystem's structural advantages (composability, developer depth, DeFi integration, cross-chain portability across L2s and EVM-compatible chains with dramatically lower gas costs) while closing the specific capability gap that makes XRPL attractive for compliance-intensive deployments.

Table 3: Cross-Ecosystem Comparison

Capability	XRPL Native	EVM Current State	This Architecture (EVM)
Multi-Purpose Tokens (fractional instruments)	Native protocol feature. Issuer creates MPT, sets transfer rules, manages compliance at issuance.	ERC-20 + ERC-3643. Functional but no binding to underlying asset.	Proposal 1: Registry-enforced asset-to-token binding. IAssetAnchorRegistry with dual-hash anchor and permanent binding. IAssetBoundToken for structural binding declaration.

Permissioned Domains	Native protocol feature. Domains define permissioned transfer environments. Transfer requires compatibility between the sender's domain and the receiver's domain.	No equivalent. Individual KYC checks cannot express directional corridor-compatibility rules between domains.	Proposal 3: ITransferDomainRegistry. Directional routes between opaque bytes32 domains by asset class. isRoutePermitted() query.
Clawback	Native protocol amendment. Issuer can clawback tokens with protocol enforcement.	ERC-3643 forcedTransfer: unstructured, any agent, any reason, no legal basis record.	Kula reference implementation: ClawbackModule snap-on. Structured trigger types, legal reference hash, timelock, appeal pathway.
Decentralised Identifiers	Native DID implementation with Verifiable Credentials.	Coupled to ONCHAINID via ERC-3643. No standard DID-agnostic interface.	Kula reference implementation: IIdentityResolver pattern. DID-agnostic, compatible with ONCHAINID, did:ethr, Polygon ID, XRPL DIDs.
Document Anchoring	Not specified at protocol level.	No standard. Bespoke implementations only.	Proposal 2: Deterministic document hashing with normalisation pipeline. Ordered concatenation bundle hashing with typed document entries.
Impact Data	Not specified at protocol level.	No standard. Off-chain PDFs.	Proposal 5: Append-only impact log with methodology attestation and independent attestor verification.
NAV / Valuation	Not specified at protocol level.	No standard. Every project implements bespoke NAV feeds via custom oracles.	Proposal 6: INAVSnapshotOracle interface with currency-keyed historical snapshots, navBasis, valuationTimestamp, and dual staleness via latestNAVStatus(). Corporate actions deferred to a separate standard.
Compliance Events	Not specified at protocol level.	No standard. VASPs implement proprietary	Proposal 4: Universal compliance event schema. Append-only log

	XRPL relies on off-chain VASP-to-VASP messaging.	solutions (Notabene, Sygna, TRP). No on-chain metadata standard.	covering full lifecycle: issuance, transfer, freeze, KYC, regulatory hold, forced transfer. Travel Rule is one event type. Queryable by regulators and auditors.
Governance	Not specified at protocol level.	Various governor frameworks, not RWA-specific.	Kula reference implementation: Two-tier governance socket. Protocol-level (PIP) and cell-level (CIP) independently deployable.
Composability	XRPL ecosystem only. Limited cross-chain.	Full EVM composability. ERC-7208-compatible data layer (optional).	Full EVM composability. ERC-7208-compatible data layer (optional). Cross-chain via existing bridge standards.

The comparison reveals a structural asymmetry. XRPL provides four of the nine capabilities natively at the protocol layer: fractional instruments, permissioned domains, clawback, and decentralised identifiers. EVM provides none of them natively. XRPL's advantage is enforcement certainty - protocol-native features cannot be circumvented by application-level code. EVM's advantage is composability - standards can be combined, extended, and upgraded without protocol changes. The six ERC proposals standardise selected capabilities that XRPL provides natively (notably fractional binding via Proposal 1 and permissioned domains via Proposal 3) and extend EVM with document anchoring, compliance records, impact reporting, and NAV interfaces that neither ecosystem standardises today. Clawback structuring, DID-agnostic identity resolution, and on-chain governance appear in the Kula reference implementation but are not part of the six normative ERC proposals.

The capabilities XRPL does not address at the protocol level include document anchoring, impact attestation, NAV reporting, and compliance event recording, are where this EVM architecture extends beyond both ecosystems' current state.

6. The Architectural Response: Six Composable ERC Proposals

The six ERC proposals are designed as independent, composable standards. Each addresses a specific problem that no published ERC standard surveyed provides. The composition is deliberate: each Proposal references the others where compositional benefit exists, but each can be adopted independently.

Table 4: Six Composable Proposals

Proposal	Title	Core Problem	Key Primitive
Proposal 1	Asset Anchor Registry Interface	No structural binding between compliant tokens and underlying titled assets	Registry-enforced asset-to-token binding with dual hash anchor, permanent binding immutability, and per-token support for ERC-721/1155.
Proposal 2	Canonical Document Bundle Anchor	No deterministic, verifiable document commitment standard	Canonical manifest format with normalization profile constants and naming convention. Schema-versioned bundle hash. Conservative profiles for JSON (RFC 8785) and XML (W3C C14N 1.1). PDF normalization deferred to companion profiles.
Proposal 3	Directional Transfer Domain Registry	No standard directional corridor-compatibility interface on EVM (XRPL has this natively)	<code>ITransferDomainRegistry</code> with <code>isRoutePermitted()</code> , <code>IGracefulRouteRevocation</code> extension. Opaque bytes32 domains, directional routes, split evidence hashes (<code>permissionEvidenceHash</code> / <code>revocationEvidenceHash</code>).
Proposal 4	Subject Linked Compliance Event Log	No on-chain standard for structured, queryable compliance events across the asset lifecycle	<code>IComplianceEventLog</code> with <code>Party</code> struct (address + role), domain-separated event type IDs, versioned payload profiles (<code>payloadProfileId</code>), correction provenance via <code>correctsIndex</code> , operation reference linking to underlying transactions,

			temporal integrity rules (occurredAt in recordEvent).
Proposal 5	Subject Linked Impact Snapshot Log	No structured, auditable, append-only impact reporting standard	IImpactSnapshotLog core with IImpactAttestation and IMethodologyVersioning extensions. Per-subject indexing, fork-preventing correction chains (correctedByIndex), domain-separated indicator IDs, indicatorSnapshotAt(), currentSnapshotForPeriod(), canonical unit encoding.
Proposal 6	Subject Linked NAV Snapshot Oracle	No standard interface for net asset value reporting on-chain	INAVSnapshotOracle core with INAVAggregation extension. Currency-keyed streams, navBasis (PER_UNIT/PER_SHARE/TOTAL), valuationTimestamp, latestNAVStatus() with dual staleness, NAV corrections via correctsIndex, domain-separated currency encoding. Corporate actions removed to separate standard.

6.1 Proposal 1: Asset Anchor Registry Interface

The core primitive. Proposal 1 defines the structural relationship between a token and the asset it represents. It introduces two coupled contract types:

TitleCell is the Kula reference implementation of the IAssetAnchorRegistry anchor record. It stores legalHash (a keccak256 commitment to the legal basis document), evidenceHash (a keccak256 commitment to the custody or evidence document), jurisdiction (ISO 3166-1 alpha-2 code), assetClass, and attestation metadata including expiresAt. TitleCell is one implementation pattern for a Proposal 1 anchor record and is not mandated by the standard.

FractionalInstrument is the Kula reference implementation of the IAssetBoundToken interface. It is an ERC-3643-compliant ERC-20 token that issues proportional economic claims against a TitleCell anchor. Each instrument declares its binding to a registry anchor via anchorId() and anchorRegistry(). Once bound, the binding is immutable. The instrument inherits compliance from ERC-3643 and extends the transfer check sequence to include domain membership verification via Proposal 3. FractionalInstrument is one implementation pattern for a Proposal 1 asset-bound token and is not mandated by the standard.

The mutual binding model is the key design decision. The registry enforces that each anchor can be bound to at most one token contract (or one specific tokenId for ERC-721 and ERC-1155 tokens), and that binding is permanent. The allocation integrity invariant - that total fractional claims cannot exceed 100% of the underlying asset - is an implementation concern enforced in the token contract itself rather than at the registry level, allowing the standard to support both fungible fractional instruments and single-instance non-fungible claims without forcing a single enforcement model on both.

The standards survey in Section 4 identifies no existing ERC standard that provides this exact combination of structural separation, mutual authorisation, and registry enforced binding exclusively within a single registry instance. ERC-3643 and CMTAT represent the instrument without specifying what it is bound to at the contract level. ERC-721, ERC-6065, and ERC-6956 represent the asset without providing a fractionalisation mechanism. ERC-4626 fractionalises vault deposits without any concept of legal title, jurisdiction, or custodian. ERC-7943 standardises compliance interfaces without any asset binding. No composition of these standards as currently specified produces the structural separation of asset identity from economic claim with mutual binding and verification that Proposal 1 proposes. Fractional allocation integrity (total claims not exceeding 100% of the underlying asset) remains a token-contract enforcement concern where the registry does not cap supply or prevent the same physical asset from being anchored in independent registries. The wrapper insight itself,

converting an indivisible property problem into a contract-level holding vehicle that issues divisible claims, is well established in legal practice through SPVs, trusts, and holding companies. What is proposed here is its expression as an ERC standard with on-chain enforcement of the binding and the allocation constraint.

The Kula reference implementation of Proposal 1 includes three snap-on modules that attach to asset anchors via a registration mechanism. These are implementation patterns rather than requirements of the standard:

- **ClawbackModule:** structured regulatory enforcement with defined trigger types, legal reference hashes, timelocks, and appeal pathways. Extends ERC-3643's unstructured forcedTransfer into an institutionally defensible enforcement mechanism.
- **IdentityResolver:** a DID-agnostic identity interface compatible with ONCHAINID, did:ethr, Polygon ID, and other W3C DID methods without prescribing the identity backend.
- The Kula reference implementation also includes an optional two-tier governance socket. Protocol Improvement Proposals (PIPs) govern platform-wide parameters at Tier 1. Asset Improvement Proposals govern decisions specific to a single anchored asset at Tier 2. Vote weight is proportional to token holdings bound to that anchor, ensuring governance power aligns with economic interest. Both tiers support manual and encoded execution modes. The governance socket is implemented as a snap-on module and is not prescribed by the Proposal standard. Not every deployment requires on-chain governance.

6.2 Proposal 2: Canonical Document Bundle Anchor

Proposal 2 solves a problem that appears simple but is technically demanding: deterministic, verifiable commitment to off-chain documents on-chain. The core requirement is that two independent parties, given the same set of documents, must always produce the same hash. This is the property that makes document verification trustless: a counterparty can hash the documents they hold and compare against the on-chain commitment without trusting the party that created it.

The standard defines a canonical manifest format and on-chain anchoring interface for document bundle commitments. It specifies how a set of off-chain documents is described via structured Document Entries, ordered deterministically, and hashed to produce a single bytes³² commitment that can be stored on-chain. The architecture separates two distinct problems that the original design combined. Manifesting - the structure, ordering, and hash derivation of a document bundle - is stable and generic and is standardised here. Normalisation - how raw files are transformed into canonical byte representations - is format-specific and evolving and is handled through versioned normalization profiles that compose with the manifest standard without requiring changes to the core ERC. The standard ships with three conservative normalization profiles: raw bytes as the universal fallback, RFC 8785 for JSON, and W3C Canonical XML 1.1 for XML. PDF and image normalization profiles are intentionally deferred to companion specifications pending binary fixtures and expert review. Until companion profiles are published, PDF and image documents SHOULD use raw-byte mode. The bundle hash computation includes a schema version prefix for domain separation, ensuring that future schema revisions cannot produce hash collisions with existing bundles. Timestamps are excluded from the Document Entry and from the hash computation. The anchoring timestamp comes from `block.timestamp` in the on-chain `AnchorRecord` and is therefore non-manipulable.

Where this standard is deployed alongside Proposal 1, the asset anchor's `legalHash` and `evidenceHash` fields are naturally produced using Proposal 2's manifest and hashing schema. The `LEGAL_BASIS` bundle produces `legalHash`; the `EVIDENCE` bundle produces `evidenceHash`. This creates a verifiable chain from the on-chain asset record to the off-chain legal documents that anchor it. This composition is optional rather than required. Proposal 2 operates as a standalone standard regardless of whether Proposal 1 is deployed, using a generic `subjectId` to define the document namespace.

Proposal 2 is deliberately scoped as a general-purpose standard. While its primary consumer in this architecture is Proposal 1, the normalisation pipeline works for any use case requiring deterministic document commitment: notarisation contracts, compliance attestation systems,

legal tech applications, ESG certification anchoring. This broader applicability strengthens its case as an independent ERC.

6.3 Proposal 3: Directional Transfer Domain Registry

Proposal 3 solves a problem that XRPL addresses at the protocol layer and EVM has no equivalent for: directional transfer compatibility between bounded domains. The question is not whether address A is allowed to hold a token (that is KYC, handled by ERC-3643). The question is whether a transfer from domain A to domain B is permitted for asset class C. The registry answers a single deterministic question: given a source domain, a destination domain, and an asset class, is the route permitted?

The `ITransferDomainRegistry` interface records directional transfer compatibility between opaque domain identifiers. A domain is an opaque `bytes32` identifier representing any logical boundary within which assets are managed, a jurisdiction, a gaming economy, an enterprise network, a DAO treasury, or any other bounded context. The registry makes no assumptions about what a domain represents; it stores only the compatibility mapping. Routes are directional: a route from domain A to domain B for asset class C does not imply a route from B to A. Bidirectional compatibility is registered as two separate routes. The core query function `isRoutePermitted()` is deterministic for any given block and does not depend on `msg.sender` or `tx.origin`. `setRoute()` only enables routes; there is no permitted boolean parameter, and disabling a route is exclusively through `revokeRoute()`. The interface uses split evidence hashes: `permissionEvidenceHash` records the justification for enabling a route (an MOU, agreement, or governance vote), while `revocationEvidenceHash` records the justification for disabling it (a regulatory order, expired agreement, or sanctions). These are separate fields so that permission evidence and revocation evidence remain independently readable in any state. `getRoute()` returns the full route state for inspection.

Graceful revocation is specified as a separate `IGracefulRouteRevocation` extension rather than being embedded in the base interface, so implementations that need only immediate revocation implement just the core. The extension adds a three-step lifecycle. `initiateRevocation()` records the revocation intent and emits an event with a computed `effectiveAt` timestamp; the route remains permitted during the grace period. Once `block.timestamp` reaches `effectiveAt`, the route is functionally revoked whether or not `finalizeRevocation()` has been called. `cancelRevocation()` reverts the pending revocation during the grace period, carrying its own `cancellationEvidenceHash` documenting the reason, and reverts if no revocation is pending or the grace period has expired. This makes the full initiate-cancel-finalize lifecycle explicit and auditable on-chain, and lets implementations operating under regulated frameworks align the grace period with the relevant settlement cycle. The mechanism for authorising registrars is an implementation choice: the standard does not specify whether the registry is permissioned or permissionless.

The finalized boolean distinguishes between in-grace-period revocations (finalized = false) and permanent revocations (finalized = true).

The domain registry is the directional compliance layer that individual KYC checks cannot provide. It verifies that a transfer from the sender's domain to the receiver's domain is permitted for the asset class before the transfer proceeds. Asset classes in this context are categories, not individual asset identifiers. Two distinct assets may share the same asset class if the same transfer rules apply to both. The derivation of asset class identifiers from individual asset records is an application-layer decision, allowing registries to operate at regulatory-category granularity rather than per-asset granularity and scale accordingly.

6.4 Proposal 4: Subject Linked Compliance Event Log

Proposal 4 defines a standard schema for emitting and querying structured compliance events across the full lifecycle of an on-chain asset. Every regulated tokenised asset generates a compliance history: tokens are issued, identities are verified, addresses are frozen, regulatory holds are placed, forced transfers are executed, KYC statuses change. Today, this history is scattered across heterogeneous event signatures, proprietary log formats, and off-chain databases. A regulator auditing a tokenised fund must parse Transfer events, custom Freeze events, off-chain KYC records, and platform-specific logs, then manually correlate them into a coherent timeline. There is no standard way to query the full compliance history of an on-chain subject across contracts, platforms, or asset classes.

The IComplianceEventLog interface defines a universal ComplianceEvent schema that captures what happened, to which subject, by whom, under what authority, with what evidence, and at what time. Each event records a subjectId and optional subjectType for namespace context, a domain-separated eventType, an outcome, the actor who performed the action, and the claimed regulatory authority. Involved addresses are recorded as an array of Party structs, each pairing an address with a domain-separated role (SENDER, RECEIVER, TARGET, BENEFICIARY, CONTROLLER, SUBJECT) rather than as a raw address list, so consumers can interpret each participant's function without out-of-band knowledge. Event-type-specific data travels in a payload field tagged by a versioned payloadProfileId (PAYLOAD_TRANSFER_V1, PAYLOAD_FREEZE_V1, PAYLOAD_KYC_V1, PAYLOAD_FORCED_TRANSFER_V1, and custom profiles), and consumers must check the profile ID before decoding, treating unrecognised profiles as opaque bytes. An operationRef field links each compliance event to the underlying on-chain transaction it documents, the recommended derivation being a hash of the transaction hash and log index, so the record is verifiably tied to the action it describes. The event type taxonomy covers the full lifecycle: issuance, transfer, redemption, freeze, unfreeze, KYC approval and revocation, regulatory hold, forced transfer, allowlist changes, policy changes, an explicit CORRECTION type, and custom types. Travel Rule compliance is one event type within this taxonomy, not the scope of the standard. Per-transfer compliance outcomes belong in the TRANSFER payload profile rather than as discrete COMPLIANCE_CHECK events, which are scoped to deliberate actions such as periodic sanctions screening.

Correction consistency rule: indicatorId, periodStart, and periodEnd must match the original event being corrected.

Append-only semantics are the core integrity guarantee. Once recorded, a compliance event cannot be modified or deleted. Corrections are new events carrying a correctsIndex referencing the original, with a NO_CORRECTION sentinel marking originals; the target's correctedByIndex is

set when it is corrected, and `recordEvent` reverts if the target has already been corrected, preventing fork chains. Temporal integrity is enforced at the contract level: `occurredAt` records when the action happened and must not be later than `block.timestamp`, `recordedAt` captures the block timestamp of recording, and implementations should restrict excessive backdating. Because a public chain makes all this permanently readable, the standard provides privacy guidance for sensitive deployments: party commitments, redact payloads, and generalise outcomes, exchanging full data off-chain. For regulated tokenised assets, the result is that the complete compliance history of any subject is permanently and verifiably on-chain, queryable by regulators and auditors without access to any proprietary off-chain system. ERC-165 interface detection is recommended.

The composition with Proposal 3 is direct. The domain registry determines whether a transfer is jurisdictionally permissible. Proposal 4 records the compliance event that documents the outcome. Together they provide both the permission layer and the compliance evidence layer. Every VASP, every regulated exchange, and every institutional custody provider operating under any regulatory framework generates compliance events of the types this standard covers. Proposal 4 has the broadest adoption surface of the six proposals precisely because it is architecture-agnostic. An ERC-3643 deployment can emit compliance events through it. A CMTAT deployment can too. The event schema is a reporting layer, not an enforcement layer.

Proposal 4 is not RWA-specific. It applies to any regulated token contract across any asset class. Demand for structured on-chain compliance event infrastructure exists across the entire regulated crypto ecosystem, not only in RWA deployments.

6.5 Proposal 5: Subject Linked Impact Snapshot Log

Proposal 5 addresses the absence of structured, auditable impact data infrastructure on-chain. Every existing RWA token standard treats sustainability and impact data as optional off-chain metadata, with no on-chain mechanism to detect silent revision of historical figures or selective disclosure. Proposal 5 provides tamper-evident impact reporting: an append-only on-chain record where the original claim, every correction, and every attestation are permanently preserved and verifiable.

The standard is structured as three interfaces: `IImpactSnapshotLog` as the core append-only log, `IImpactAttestation` as an optional attestation extension, and `IMethodologyVersioning` as an optional methodology-evolution extension. The core log attaches per-subject indexed snapshots to a subject identifier. Each `IndicatorSnapshot` records a quantitative measurement for a specific indicator over a measurement period expressed as a half-open interval, along with a methodology hash and URI. Indicator identifiers are domain-separated, derived as `keccak256("Proposal-XXXX:INDICATOR:CARBON_OFFSET")` and similar, and units use canonical encodings such as `keccak256("tCO2e")` or `keccak256("kWh")` so that measurements are unambiguous across implementations. The methodology declaration is mandatory: no number exists without its method, and the first snapshot recorded for a given subject and indicator initialises the active methodology. Retrieval functions include `indicatorSnapshotAt()` for per-indicator access by ordinal and `currentSnapshotForPeriod()` for the current value covering a specific period. `latestIndicatorSnapshot()` returns the latest snapshot by submission time, which is not necessarily the latest correction of a specific period, a distinction consumers must account for.

The attestation extension enables attestors to verify reported metrics and record their assessment on-chain. An attestor can endorse or dispute a snapshot, providing an evidence hash and a stored, retrievable `evidenceURI`, with attestations retrieved by ordinal index through `attestationCount()` and `getAttestation()`. This creates a two-layer audit trail: the reporter and attestor each leave an independent record. The standard refers to attestors generically rather than as third parties; implementations operating under regulated frameworks should restrict attestation to credentialed attestors, since unrestricted attestation reduces the evidentiary weight of endorsements. The methodology-versioning extension records methodology evolution over time, with supersession constrained to take effect only from a future ordinal so that historical snapshots retain the methodology under which they were reported. ERC-165 interface detection is recommended for all three interfaces.

Append-only semantics are the core integrity guarantee. Once recorded, a snapshot cannot be modified or deleted. Corrections are issued as new snapshots with a `correctsIndex` field that

creates an explicit on-chain provenance chain. Original measurements use `type(uint256).max` as the sentinel value for `correctsIndex`, unambiguously marking the snapshot as a first-time report and avoiding collision with snapshot index 0, which is a valid index for the first recorded snapshot. Corrections set `correctsIndex` to the `snapshotIndex` of the snapshot being corrected, creating a linked list traversable from the latest value back through every revision to the original claim. The original claim, the correction, and the attestation history are all permanently on-chain. The standard also addresses methodology evolution. When a regulatory framework such as SFDR is revised, historical data reported under the prior framework does not become invalid, but it needs to be re-contextualised. The `supersedeMethodology` function records a framework transition as a first-class event, linking the old methodology to the new and identifying from which snapshot the new methodology applies. Historical snapshots retain their original methodology references. This makes methodology transitions explicit and auditable rather than creating a silent discontinuity in the data series.

The indicator taxonomy defines base identifiers (`CARBON_OFFSET`, `ENERGY_GENERATED`, `JOBS_CREATED`, `BIODIVERSITY_AREA`, and others) using domain-separated derivation. Fork prevention applies to the correction chain through `correctedByIndex`: each snapshot can be corrected at most once, so the provenance chain cannot branch. Custom indicators use the same derivation pattern with mandatory documentation requirements.

6.6 Proposal 6: Subject Linked NAV Snapshot Oracle

Every tokenised real-world asset has a net asset value. For DeFi protocols to use RWA tokens as collateral, for investors to evaluate holdings, and for fund administrators to calculate portfolio-level NAV, there must be a standardised way to report asset valuation on-chain. Today, every project implements this bespoke: custom oracle integrations, proprietary NAV feed contracts, one-off event schemas. There is no standard interface that any protocol can query to retrieve the current or historical valuation of an on-chain asset.

Proposal 6 defines the `INAVSnapshotOracle` interface for publishing Net Asset Value snapshots attached to a subject identifier, with an optional `INAVAggregation` extension. Every NAV stream is keyed by subject identifier and currency, so an asset can carry independent valuations in multiple denominations. Each snapshot records the reported value with explicit decimal semantics (the value scaled by ten to the negative decimals, decimals bounded between zero and eighteen), a `navBasis` field distinguishing `PER_UNIT`, `PER_SHARE`, and `TOTAL` valuations through domain-separated constants, a `valuationTimestamp` expressed as a Unix timestamp rather than a calendar date, the valuation methodology, a methodology hash (may use Proposal 2 bundle hash for multi-document valuation memos), and the reporting provider. Snapshots are append-only, following the same integrity model as Proposal 5: NAV corrections carry a `correctsIndex` and set the target `correctedByIndex` with the same fork prevention used across the suite, so the correction chain cannot branch. The `latestNAVStatus()` function is a read-only view function (not pure) that returns two independent staleness booleans: `isPublishStale` (true when `block.timestamp` minus `publishedAt` exceeds `heartbeat` is exceeded, detecting missed publication updates) and `isValuationStale` (true when `valuationTimestamp` plus `maxValuationAge` is exceeded, detecting whether the asset has been recently valued regardless of publication timing). Both checks are necessary because a provider can publish a stale valuation promptly, satisfying the heartbeat while the valuation timestamp itself is outdated. Because it is a view function it emits no events; a separate `StalenessConfigUpdated` event records changes to `heartbeat` or `maxValuationAge`. ERC-4626 vaults that source share pricing from this interface MUST consult `latestNAVStatus()` for any function affecting redemption or subscription pricing, since consuming a NAV without checking staleness exposes users to stale-price risk. NAV is reported value, not executable price: the interface defines how to report and retrieve valuations, not a tradable quote. Multi-provider fixed median aggregation (`latestAggregationTimestamp`) is specified through the `INAVAggregation` extension, which groups submissions by subject, currency, and valuation timestamp, exposes bounded provider history through `providerSubmissionCount()` and `providerSubmissionAt()`, combines independent valuations into a consensus NAV using median, mean, weighted, or quorum methods, and surfaces material divergence through a

NAVDeviationDetected event and a deviationThreshold() getter. ERC-165 interface detection is recommended.

The latestAggregationTimestamp() function returns the block timestamp of the most recent aggregation, enabling staleness validation.

The latestNAV() function returns the terminal snapshot; in multi-snapshot scenarios, tie-breaker order is: publishedAt (most recent first), then snapshotIndex (highest first).

The NO_CORRECTED_BY = 0 constant indicates an event has not been superseded by a correction.

Corporate actions are deliberately out of scope for this standard. Real-world assets generate events that affect valuation and investor rights, distributions, capital calls, restructuring, maturity, and wind-down, but recording these is a distinct concern from reporting valuation. Bundling them into the NAV oracle would have widened the interface and weakened its claim to be a minimal, single-purpose standard. Corporate action recording is noted as a separate companion standard to be specified independently.

The composition with the rest of the stack is direct. Asset-bound tokens implementing Proposal 1 can reference the NAV interface to determine distribution amounts. DeFi lending protocols can query the NAV interface to set collateral ratios dynamically. Fund administrators can aggregate NAV across multiple anchorIds for portfolio-level reporting. The interface is deliberately minimal: it defines how to report and retrieve valuations, not how to calculate them. Valuation methodology is an application-layer concern that varies by asset class, jurisdiction, and regulatory framework.

6.7 Composition

The six proposals are designed as independent, composable standards. Each can be adopted without the others. A deployment that needs only NAV reporting can adopt Proposal 6 without deploying Proposal 1 through 4. A deployment that needs only document anchoring can adopt Proposal 2 as a general-purpose standard independent of any asset registry. The architectural value compounds when multiple Proposals are deployed together, but no Proposal requires any other as a dependency.

Composition Chain

When deployed together, the six Proposals form a coherent architecture. Proposal 1 identifies the asset and binds instruments to it. Proposal 3 determines where those instruments can be transferred. Proposal 2 verifies the documents that anchor the asset's legal and evidentiary basis. Proposal 5 records the impact the asset generates. Proposal 6 reports the asset's valuation. Proposal 4 records the compliance events across the asset's lifecycle. Where a deployment uses Proposal 1, the anchorId it defines provides a common reference that other Proposals can optionally attach to. Where Proposals are deployed independently, they operate as standalone standards with their own subject identifiers.

The stack sits on top of existing standards rather than replacing them. ERC-3643 provides the compliance engine. ERC-20 provides fungible token mechanics. ERC-7943, now final, provides the external compliance interface. ERC-7208 offers a composable data layer the suite can adopt where a deployment wants it, though the reference implementations do not require it and store state directly. The six new proposals fill the gaps that those standards leave open.

6.8 Minimal Adoption Path

Teams do not need to adopt all six Proposals to get value from the family. Proposal 2 (Canonical Document Bundle Anchor) and Proposal 4 (Compliance Event Log) can be deployed independently against any existing token standard today with minimal integration friction. Proposal 6 (NAV Oracle Interface) can be added to any ERC-4626 vault or custom NAV contract as a thin interface layer. These three are the lowest-friction entry points. Proposal 1 (Asset Anchor Registry Interface) is the right starting point for teams doing net-new titled asset deployments where the binding architecture can be established from the outset rather than retrofitted. Proposal 3 adds directional transfer-domain routing when a deployment requires it, Proposal 5 adds impact data infrastructure and Proposals 2 and 4 add document anchoring and

compliance event recording respectively. The standards are designed to be adopted incrementally, not as an all-or-nothing stack.

Table 5: Implementation Status

Proposal	Title	Spec Status	Ref. Implementation	Audit / Testnet
Proposal 1	Asset Anchor Registry Interface	Pre-submission internal review	Developed	Audited
Proposal 2	Canonical Document Bundle Anchor	Pre-submission internal review	Developed	Audited
Proposal 3	Directional Transfer Domain Registry	Pre-submission internal review	Developed	Audited
Proposal 4	Subject Linked Compliance Event Log	Pre-submission internal review	Developed	Audited
Proposal 5	Subject Linked Impact Snapshot Log	Pre-submission internal review	Developed	Audited
Proposal 6	Subject Linked NAV Snapshot Oracle	Pre-submission internal review	Developed	Audited

7. Deployment Economics

A standard architecture is only viable if the gas costs of deployment and operation are proportionate to the asset sizes it serves. This section provides gas cost analysis for the full stack on Avalanche C-Chain, which is one target deployment environment for reference implementation.

Table 6: Gas Costs

Operation	Estimated Gas	Cost at 25 nAVAX, AVAX \$9.40	Frequency
Full stack deployment (asset anchor + bound token + modules)	~3,000,000	~\$2.89	Once per asset
Compliant transfer (five- step check)	~65,000	~\$0.02	Per transfer
PAI indicator snapshot	~150,000	~\$0.05	Per reporting period
Document hash anchoring	~120,000	~\$0.04	Per document update
Domain compatibility declaration	~80,000	~\$0.03	Per domain pair
Clawback request	~180,000	~\$0.06	Rare (enforcement action)
NAV snapshot publication	~130,000	~\$0.04	Monthly or quarterly
Corporate action event (companion standard)	~100,000	~\$0.03	Per event
Compliance event (per transfer)	~90,000	~\$0.03	Per regulated transfer

Full year monthly PAI reporting (12 indicators)	~21,600,000	~\$7.20	Annual
Full year monthly NAV (corporate actions: companion standard)	~1,960,000	~\$0.65	Annual

These economics make the architecture viable at emerging market asset sizes (\$3-50 million) rather than requiring institutional scale (\$500 million and above) to justify the overhead. A \$5 million Tungsten concession, a \$10 million solar installation, or a \$20 million real estate development can deploy the full stack for under \$3 and operate it for under \$10 per year in impact reporting costs.

The gas cost profile also reveals a design advantage of the modular architecture. The expensive operation (deployment) happens once per asset, while the frequent operations (transfers, attestations) are individually cheap. This is the correct cost distribution for illiquid real-world assets that trade infrequently but report regularly.

On Ethereum mainnet, the same operations would cost approximately 100-200x more at current gas prices, making the architecture viable only for larger asset sizes. Layer 2 solutions (Arbitrum, Optimism, Base) offer intermediate cost points. The standards themselves are chain-agnostic so the deployment economics are an implementation consideration. The choice of Avalanche as one target environment is validated by institutional precedent. BridgeTower Capital's \$11 billion DOM X copper-gold tokenisation, one of the largest single-asset deployments to date, was issued on Avalanche, confirming institutional confidence in the chain's cost profile, finality characteristics, and regulatory positioning for natural resource and infrastructure assets.

8. The Case for Standards Over Bespoke Infrastructure

The six ERC Proposals presented in this paper invite a reasonable objection that if everything they specify can already be built without them, what problem do they solve?

The answer is straightforward. The capability to build all this already exists. BridgeTower succeeded with four bespoke Chainlink components. Ctrl Alt built custom XRPL infrastructure. AMINA, Tokeny, and 21X assembled a complete Swiss securities pathway. Every capability that the six Proposals specify exists in production somewhere. The question is not whether these things can be built. The question is whether their absence as standards forces every participant to build the same things independently, and whether that creates compounding fragmentation as the ecosystem grows. The jurisdictional evidence in Section 1.1 answers that question: Dubai, Switzerland, Germany, and BridgeTower are building production infrastructure for titled asset tokenisation. None of it interoperates. Each new entrant rebuilds from scratch or adopts one project's stack with the vendor dependency that implies.

8.1 Risks to Adoption

These Proposals face real adoption risks. The EIP Proposal process can be slow. ERC-3643 took over two years with a dedicated commercial sponsor(s) behind it. Teams with working audited deployments have limited incentive to adopt new unproven interfaces. Audit costs are non-trivial. Multi-chain market fragmentation narrows the reach of an EVM-only suite. None of this invalidates the proposals. It argues for careful sequencing, prioritising the highest-applicability Proposals first, building reference implementations before submission, and engaging ecosystem players as co-submitters.

When Dubai's tokenised deed tokens need to be accepted as collateral by a DeFi lending protocol built on Ethereum, the protocol needs to verify the binding between the token and the underlying asset. If Dubai's binding mechanism is bespoke, that verification requires a custom integration. When BridgeTower's NAV data needs to be read by a secondary market operator, the operator needs to know what interface to call. If BridgeTower's NAV feed is a proprietary Chainlink NAVLink integration, every consumer of that data builds a bespoke adapter. When a regulated transfer between a Mauritius-domiciled instrument and a Luxembourg-domiciled counterparty needs Travel Rule compliance, both the originating and receiving VASPs need to verify that compliance occurred. If the compliance attestation sits only in Notabene's off-chain messaging system, on-chain verification requires trusting Notabene's platform rather than reading a standardised on-chain record.

The history of Ethereum token standards provides the precedent. ERC-20 did not make fungible tokens possible, fungible tokens existed before ERC-20. What ERC-20 made possible was composability. Any wallet could read any token's balance, any exchange could list any token, any lending protocol could accept any token as collateral, without custom integration. That single interface created the conditions for decentralised finance to emerge. Not the token itself. The standard interface.

The six proposals presented here operate at the next layer of the same progression. Proposal 1 does not invent asset-to-token binding, BridgeTower has it, Ctrl Alt has it. It standardises the interface so that any secondary market, lending protocol, or compliance system can verify the binding without trusting the issuer's bespoke implementation. Proposal 3 does not invent jurisdictional transfer compliance, XRPL has it natively. It standardises the interface on EVM so that corridor compatibility can be verified by any counterparty. Proposal 2 does not invent document hashing. It standardises the normalisation pipeline so that two independent parties always produce the same hash from the same documents. Proposal 5 does not invent impact reporting. It standardises the on-chain schema so that impact data is auditable, append-only, and comparable across assets. Proposal 6 does not invent NAV reporting. It standardises the interface so that any protocol can query any asset's valuation without a custom integration. Proposal 4 does not invent compliance record-keeping. It standardises the event schema so that regulators, auditors, and counterparties can query the full compliance history of any on-chain subject without accessing a proprietary off-chain system.

Open standards eliminate that dependency. They allow new entrants to build against a published interface rather than a proprietary implementation. They allow existing providers to enhance their services by anchoring them to a common layer. They allow regulators to reference a standard rather than evaluating each bespoke deployment individually.

The six Proposals provide the interface layer that composability requires. Whether that layer gets adopted depends on implementation quality, community engagement, and the decisions of projects building in this space. This paper makes the case for why the interfaces are necessary. The reference implementations and Proposal submissions that follow will determine whether they are used.

9. Conclusion

Existing ERC standards solve their individual problems well. What the standards survey in Section 4 does not identify is the layer that connects a compliant fractional instrument to a specific underlying titled asset in a structurally enforced, directionally constrained, verifiable through document commitments, and auditable and measurable way. The six Proposals address that gap without replacing any existing standard. Each extends the ecosystem into territory the current landscape leaves unaddressed.

The theoretical grounding for this work is drawn from the tokenisation taxonomy “Where Authority Resides” (Turner, 2026), which identifies the boundary between contractual and title tokenisation as the critical transition in the progression toward internalised completion authority. The legal and institutional dimensions of that boundary are documented in the DLT securities literature, including Zetsche et al. (2020) on decentralised finance and regulatory treatment and Micheler and Whaley (2019) on regulatory technology and code as law. The six Proposals occupy that boundary technically. They do not claim to achieve title tokenisation by themselves. That requires statutory recognition, which is a legislative achievement, not a technical one.

The regulatory trajectory described in Section 3 is moving toward requiring the capabilities these Proposals provide. The technical infrastructure those frameworks assume does not yet exist as published Ethereum standards.

The Proposal submissions are open source (CC0). The standards are the contribution. Any implementation is one instance of them.

Formal Proposal submissions, audited reference implementations, and public testnet deployments will follow this paper. Kula intends to engage directly with the Ethereum Magicians community, the ERC-3643 Association, and existing RWA infrastructure providers to ensure the proposals are positioned as complementary extensions rather than competing alternatives.

References

BridgeTower Capital. From Chainlink Early Adopter to Live CRE Deployment: How BridgeTower Built a Production Tokenization Platform for an \$11B+ Arizona Copper-Gold Asset. Press Release, April 2026.

CMTAT: Capital Markets and Technology Association Token Standard. <https://cmtat.github.io/>

ERC-20: Token Standard. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-20>

ERC-721: Non-Fungible Token Standard. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-721>

ERC-1155: Multi Token Standard. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-1155>

ERC-3643: T-REX - Token for Regulated Exchanges. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-3643>

ERC-4626: Tokenized Vaults. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-4626>

ERC-6065: Real Estate Token. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-6065>

ERC-6956: Asset-bound Non-Fungible Tokens. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-6956>

ERC-7208: On-Chain Data Container. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-7208>

ERC-7518: Dynamic Compliance Interop Security Token (DyCIST). Ethereum Improvement Proposals. <https://eips.ethereum.org/EIPS/eip-7518>

ERC-7551: Crypto Security Token Smart Contract Interface (eWpG). Ethereum Improvement Proposals. <https://eips.ethereum.org/EIPS/eip-7551>

ERC-7929: PermaLink Asset Bound Token. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-7929>

ERC-7943: Universal RWA Interface (uRWA). Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-7943>

ERC-7540: Asynchronous ERC-4626 Tokenized Vaults. Ethereum Improvement Proposals. <https://eips.ethereum.org/EIPS/eip-7540>

ERC-7575: Multi-Asset ERC-4626 Vaults. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-7575>

ERC-7726: Common Quote Oracle. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-7726>

ERC-7802: Crosschain Token Interface. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-7802>

ERC-8106: RWA Event-based Compliance Framework. Ethereum Improvement Proposals. <https://eips.ethereum.org/EIPS/eip-8106>

ERC-8127: Human Readable Token Identifiers. Ethereum Improvement Proposals.
<https://eips.ethereum.org/EIPS/eip-8127>

Ethereum Attestation Service (EAS). <https://attest.sh/>

EU Transfer of Funds Regulation (TFR): Regulation (EU) 2023/1113 on information accompanying transfers of funds and certain crypto assets.

FATF Recommendation 16: Wire Transfers (The Travel Rule). Financial Action Task Force, updated 2023.

German eWpG: Gesetz über elektronische Wertpapiere, 2021.

Hypercerts. <https://hypercerts.org/>

ISSB IFRS S2: Climate-related Disclosures. International Sustainability Standards Board, 2023.

Micheler, E. and Whaley, A. (2019). Regulatory Technology: Replacing Law with Computer Code. European Business Organization Law Review, 21(3), 349-377.

Notabene Travel Rule Protocol. <https://notabene.id/>

OpenVASP: Open Virtual Assets Service Provider Protocol. <https://openvasp.org/>

SFDR: Regulation (EU) 2019/2088 on sustainability-related disclosures in the financial services sector.

Swiss DLT Act: Federal Act on the Adaptation of Federal Law to Developments in Distributed Ledger Technology, 2021.

Turner, C. (2026). Where Authority Resides: A Completion-Based Taxonomy of Tokenisation. SSRN Working Paper.

XRPL Clawback Amendment. <https://xrpl.org/docs/concepts/tokens/clawback>

XRPL Decentralized Identifiers. <https://xrpl.org/docs/concepts/accounts/decentralized-identifiers>

XRPL Multi-Purpose Tokens. <https://xrpl.org/docs/concepts/tokens/multi-purpose-tokens>

XRPL Permissioned Domains. <https://xrpl.org/docs/concepts/accounts/permissioned-domains>

Zetsche, D., Arner, D., and Buckley, R. (2020). Decentralized Finance. *Journal of Financial Regulation*, 6(2), 172-203.

Annex 1 - Why these six standards?

A brief review of what each standard does, what it contributes that existing standards do not, and a grounded case for why they each matter.

This annex describes the six proposed standards one at a time. For each, it sets out what the standard does as a piece of technology, what it contributes that existing standards do not already provide (from the perspective of composability and complementarity), a grounded situation that shows the contribution in use, and what is at stake for institutional adoption if the problem goes unsolved. The situations described are drawn from genuine industry and regulatory conditions, and they are written independently of any single issuer so that each standard can be judged on its own terms.

Standard One's `registerAnchor()` produces an `anchorId`. In composed deployments, that identifier becomes the `subjectId` in Standards 2-6. This is how all six reference the same underlying asset without importing each other. Standards 2-6 remain standalone as specs; composition is a deployment convention.

The `computeAnchorId()` function enables atomic register-and-bind transactions, ensuring the anchor and token binding are established in a single operation.

These standards enable institutional-grade instruments in permissioned ecosystems. They do not create on-chain title (which requires statute legal acceptance). They make claims verifiable and portable enough for institutional capital to participate before statutory title recognition exists.

Normative detail lives in the six Proposal specifications and the Technical Architecture Paper. This annex explains the practical why.

How the Six Standards Compose

Underneath the six proposed standards there is one question which decides whether tokenisation becomes a powerful dynamic market with strong global liquidity, or stays a collection of experiments that yes, are rapidly scaling, but are being developed in siloed business models.

What does it take for institutional and sovereign capital to trust an asset deployed on-chain?

Standards 2, 4, and 6 can be adopted standalone against existing tokens. Full titled-asset composition uses Standard One as the identity root.

No single feature or tokenisation product answers this question. It is answered by whether trust can travel. Any competent team can solve its own problems with bespoke design, and many

have. But bespoke design produces trust that is local. It holds inside one platform, one issuer, one technology stack, one jurisdiction, and it has to be rebuilt every time capital crosses from one to another. A market made of bespoke implementations is only ever as strong as its weakest participant, because capital cannot easily tell the sound implementations from the unsound without examining each one, and a single failure contaminates confidence in all of them. This friction is real and is the ceiling to the potential of tokenization at scale.

A shared set of standards changes the property of the trust rather than the quality of any one implementation. It makes verification portable. A regulator who has confirmed that an asset conforms to a standard does not have to start again when the asset moves to another venue. An allocator who understands the standard understands every asset built to it. Capital can move across the whole field at low friction because the verification travels with the asset instead of staying with the platform that issued it. That portability is where wealth is built, not in any one tokenised asset but in the free movement of capital across a market whose floor holds everywhere. For sovereign capital, which moves on precedent and cannot afford to be the institution that funded the asset that was tokenised twice, a verifiable floor that travels, is closer to a precondition than a convenience.

These standards do not replace the compliance, identity, and verification services that already exist, and they are not intended to. They give those services something verifiable to anchor to. A Travel Rule provider still performs its checks. What the infrastructure adds is that the facts those checks rely on, the identity of the asset, the permissibility of the corridor, the integrity of the record, are verifiable at a layer beneath the service, so that one provider in Singapore and another in Frankfurt reason about the same secure ground truth. The standard makes the software layer globally consistent rather than redundant. It is a handshake at the infrastructural level that makes every solution built above it stronger, which is why the contribution is complementary to the market rather than competitive to it.

One discipline runs through everything that follows. These standards make facts verifiable and portable. They do not manufacture the underlying truth. A standard does not make an asset real. It makes the claim of its reality checkable by anyone without trusting the party asserting it, and it makes a contradiction, such as the same asset bound twice, detectable. Where this annex says a standard establishes a fact, it means the fact becomes independently verifiable and travels with the asset, not that the standard guarantees the world behind it. That distinction is what keeps the contribution honest, and it is the version that survives the scrutiny of the institutions the market most needs to convince.

These standards then operate at the infrastructure layer. They make what is asserted on-chain verifiable and portable across platforms. They do not, however, address the layer above them -

who asserts what, and on what authority. Every standard in this family assumes a layer of institutional governance, compliance practice, and documentation that sits above the technology. No standard can solve that problem. What the standards accomplish is making whatever facts are asserted verifiable at scale, auditable independently, and impossible to revise silently. At institutional and sovereign scale, that infrastructure for verification is the precondition for capital deployment. The institutional confidence the market needs does not come from preventing dishonest assertions, that is a governance and accountability question, and a normative practice in structing institutional financial capital. It comes from making assertions that are verifiable, permanent, and traceable across platforms so that any divergence from stated claims becomes detectable.

Asset Anchor Registry Interface

What it does

The evidenceHash commits to a Physical Custody Commitment (PCC) document - the jurisdictional trust anchor that bootstraps asset identity. The legal basis commitment (legalHash) and custody evidence commitment (evidenceHash) are produced using Standard Two's deterministic bundle hashing. This standard verifies the binding. It does not vouch for PCC truth.

Reference final spec: mutual binding (registry and token must agree); one anchor per token within a registry; metadata/expiry in extension interfaces (IAssetAnchorMetadata, IAssetAnchorRenewal, IAssetAnchorDeactivation); ERC-165 as preferred conformance signal.

This standard defines a registry that records a permanent, one-to-one binding between a token and a claim about a specific off-chain asset. The binding carries two cryptographic commitments - one to the legal basis of the asset, and one to the evidence of its existence or custody. Once a token is bound to an asset record, that binding cannot be reassigned or silently altered, and any party can query the registry to confirm what a given token is bound to. The standard expresses the binding at the level a token requires. A fungible token binds at the contract level, while a non-fungible or multi-token contract binds at the level of the individual token identifier.

What it contributes

Existing token standards describe how a token behaves. They do not describe what it represents. The relationship between a token and the asset it stands for is, in current practice, asserted in a legal document the contract never references, and a counterparty cannot independently check. This standard closes that gap. It is the first interface to make asset-to-token binding a structural, verifiable property rather than a claim in an offering memorandum, and it makes that verification portable: the binding is confirmed the same way regardless of which platform the token trades on. It also supports fractionalised deployments by making one-to-one anchor ↔ token binding within a registry verifiable on-chain. Supply caps and over-allocation prevention are enforced in the token contract (or off-chain legal structure), not by the registry. No existing standard provides this combination of permanent binding, dual legal-and-evidence commitment, and single-registry binding exclusivity.

This standard makes asset binding verifiable. Attestation layer = title registries, legal counsel, custody agents.

In use

Consider a single high-value asset, a parcel of land or an industrial facility, that an issuer wishes to open to many investors. The asset is indivisible, and its legal title is unitary, but investors want divisible, tradeable fractions. Today the link between each fraction and the underlying asset lives only in off-chain agreements, and a buyer of a fraction on a secondary market has no way to confirm, at the moment of purchase, that the fraction is genuinely bound to the asset they think it is, or that the issuer has not bound more than the whole of the asset across separate instruments. With a binding registry, a counterparty can confirm in a single on-chain query which anchor a token claims and whether the registry confirms that binding. Preventing over-allocation of fractional claims still requires token-level supply controls and diligence on cross-registry duplication; the registry alone does not make over-allocation structurally impossible.

What is at stake?

The first thing institutional and/or sovereign capital must be certain of, before any question of yield or impact, is that it owns what it believes it owns and that the same asset has not been sold out from under it elsewhere. Without a verifiable binding that travels with the token, that certainty depends on trusting each issuer's private records, and at market scale that trust does not hold. Double-counting need not even be fraudulent to be fatal to confidence. It need only be unprovable to be absent. This is the foundation the rest depends on, because every other guarantee is worthless if the token is not reliably tied to a real and singular asset in the first place.

Canonical Document Bundle Anchor

What it does

This standard defines a deterministic way to describe a set of documents, order them, and reduce them to a single commitment that can be stored on-chain. It separates two concerns that are often conflated. The structure and ordering of a document bundle, which is stable and general, and the conversion of individual files into a canonical byte form, which is format-specific and handled through versioned profiles. Two parties holding the same set of documents will independently compute the same commitment, and any later alteration to any document in the set becomes detectable.

What it contributes

a) No on-chain normalization profile constants and naming convention - only profile constants and naming; (b) Anchoring timestamp is consensus block time, not non-manipulable; (c) Records keyed by (bundleHash, subjectId, role); (d) documentCount is declarative - consumers recompute manifest off-chain; (e) C14N 1.1 is distinct from Exclusive XML Canonicalization.

Document hashing on-chain is not new. Commercial platforms already hash documents and link them to transactions, and several do it well. What they do is proprietary to each platform, and the method by which the commitment is computed is internal to it. This standard contributes the open, deterministic, cross-platform method. A way of computing a document commitment that is specified publicly, so that a commitment produced by one system can be verified by another without trusting the system that produced it. It deliberately defers the hardest format-specific cases, such as the canonical treatment of signed PDFs, to companion profiles rather than publishing a method that cannot guarantee that two correct implementations agree. The contribution is a shared standard of verification, which is precisely the property that lets verification travel between parties who share no platform.

Notarisation/issuance systems produce documents; standard verifies commitment.

In use

A pension fund conducting due diligence on a tokenised infrastructure asset is shown the legal documents behind it - the concession agreement, the custody confirmation, the audited accounts. The fund's counsel wishes to confirm these are exactly the documents the issuer committed to at issuance, with nothing substituted in the interim. If the on-chain commitment was produced by a method internal to the issuing platform, the fund must either trust that platform or reverse-engineer its method. With an open canonical method, the fund's own

systems compute the commitment independently and compare. The documents are either provably the originals or provably not.

What is at stake?

What does an institution do when the proof of an asset's legal basis can only be checked by the same platform that issued it? It either accepts platform-bound assurance, which does not satisfy a serious diligence standard, or it conducts full manual diligence on every asset, which does not scale. Both answers cap the market well below the volumes being projected for it. Portable document verification removes that ceiling by letting any party confirm the paperwork for any asset without depending on the issuer's own systems to vouch for it.

Directional Transfer Domain Registry

What it does

This standard defines a registry of directional rules governing whether a transfer between two bounded contexts is permitted for a given class of asset. A context, called a domain, is an opaque identifier that can represent a jurisdiction, an investor pool, an enterprise network, or any other boundary an issuer must reason about. Rules are directional. Permission to move from one domain to another does not imply permission to move in reverse. Each rule carries separate evidence for why it was enabled and, where applicable, why it was revoked, and an optional extension lets a revocation take effect after a grace period so that in-flight transactions are protected. The registry answers one deterministic question on every transfer. Given a source domain, a destination domain, and an asset class, is this movement permitted.

What it contributes

The honest position matters here. Existing compliance standards already control whether a particular holder may hold or receive a token, which is a property of the holder and is well solved. This standard does not duplicate that. Its contribution is twofold. First, it expresses a rule that holder-level eligibility cannot capture. A rule about the direction of movement between contexts, independent of the individuals involved, where two fully eligible parties may transact in one direction and be forbidden in the other because the rule concerns the corridor rather than the persons. Second, and more importantly at scale, it moves corridor enforcement from the application layer, where it is platform-specific and observable only to the platform that implements it, down to the infrastructural layer, where the rule travels with the asset and is readable by a regulator directly from the chain regardless of which venue the asset trades on. That shift, from many private enforcement implementations to one verifiable infrastructural guarantee, is the contribution.

Corridor regulators/MOUs sit above registry.

Protocol enforcement is impossible on a permissionless chain. Conformance is a market signal: (1) ERC-165 introspection distinguishes conforming from non-conforming contracts, (2) market participants refuse non-conforming contracts, (3) regulators observe which implementations conform.

The IGracefulRouteRevocation extension - corridor closure with a grace period for in-flight settlement - is operationally critical for regulated environments and is not addressed by any existing standard. Grace period duration is implementation-defined; only the computed effectiveAt timestamp is standardized.

In use

Any jurisdictions that permit foreign investment in strategic assets attach conditions to preserve domestic participation. A mining concession may be opened to international investors only on the condition that local ownership is not diluted below an agreed level: local holders may sell out to international buyers, but international holders may not accumulate in the reverse direction past a ceiling. Every investor on both sides may be fully verified, so a per-holder eligibility check sees nothing wrong with any single transfer. At primary issuance the condition is easy to honour, because the issuer allocates by hand. The difficulty arrives in the secondary market. Once tokens trade peer-to-peer, no human is allocating and no single intermediary sits in every transfer path, so the condition can be breached not by any one bad transfer but by the cumulative drift of ordinary trading across a corridor the regulator drew. A directional corridor enforced at the infrastructural level holds the condition automatically, on every transfer, primary or secondary, because the rule belongs to the asset rather than to the venue.

What is at stake?

This is the standard whose necessity is written into the trajectory of the market rather than its present size. At today's volumes, with most activity intermediated and secondary markets thin, platform-level corridor enforcement is sufficient and this standard is largely latent. The conditions that make it load-bearing are precisely the conditions the market is projected to reach with deep secondary markets and disintermediated transfer, where no single Travel Rule provider sits in every path to maintain line of sight. At that scale, a regulator asking whether an asset can be guaranteed never to cross a forbidden corridor cannot be answered by a patchwork of platform-specific assurances, each only as reliable as its weakest implementation. It can be answered only if the corridor rule is enforced at the infrastructural level and travels with the asset. This standard provides the corridor-permissibility half of that guarantee and composes with the identity and event-logging layers that carry the originator and beneficiary half. Without an infrastructural guarantee of this kind, cross-jurisdictional line of sight cannot be assured as the market scales, and the regulatory certainty that sovereign capital requires before deploying on tokenised rails does not exist.

Subject-Linked Compliance Event Log

What it does

This standard defines a structured, append-only record of compliance-significant events attached to a subject. Each event captures what happened, to which subject, performed by whom, under what claimed authority, with what supporting evidence, and at what time the underlying action occurred. Parties are recorded with explicit roles rather than as bare addresses, event data travels in versioned formats so that consumers can interpret it without external documentation, and corrections are made as new entries that reference the entry they correct, so the original record is never overwritten, and the correction history cannot fork.

What it contributes

KYC-sensitive data lives off-chain. On-chain events record cryptographic commitments (hashes) to that data, not the data itself. Off-chain data can be deleted (right to erasure) while the on-chain commitment remains valid as proof the event occurred.

operationRef is an application-level correlation ID set by the caller - not derived inside the contract from the transaction hash.

Compliance platforms keep audit logs today, and a separate proposal exists for event-based compliance using soft advisory signals. This standard differs in two specific respects. It is structured for permanence and attribution, so every event carries who acted and under what authority, recorded immutably, which is what distinguishes an evidentiary record from an operational log that can be edited. And it is designed to be interpretable by any party rather than only by the system that wrote it, through explicit roles and versioned payloads. The contribution is a common evidentiary schema that a regulator or auditor can read directly from the chain regardless of the platform that produced it, which is the same portability of trust applied to compliance history.

KYC/AML providers perform checks; standard records verifiable commitments.

In use

When a regulated fund is examined, the examiner does not want only the current state of the cap table, rather, they want the history of how it reached that state, who approved each change and on what authority, with assurance that the history has not been tidied before presentation. In conventional systems this assurance rests on the integrity of the record-keeper. An append-only, attributed, on-chain event log makes the history independently verifiable. The examiner reads the sequence of events, sees that corrections are additive rather than silent overwrites, and confirms that the record presented is the record as it always was.

What is at stake?

Institutional and sovereign allocators are themselves examined by their own regulators, and they must be able to demonstrate the compliance history of every position they hold. If that history can be reconstructed only from the records of the platform that maintained it, the allocator inherits that platform as a single point of failure in its own regulatory standing. Few serious institutions will accept that exposure. A history that is permanent, attributed, and readable independently of any platform turns regulatory reporting from a dependency on a vendor into a fact the institution can stand behind in front of its own supervisor. Regulators that require durable audit trails include MiCA Art. 68 + RTS, VARA retention, SEC 17a-3/17a-4, FCA consultations.

Subject-Linked Impact Snapshot Log

What it does

This standard defines an append-only record of measured impact data attached to a subject. Each entry records a quantitative measurement of a specific indicator over a defined period, together with the methodology used and a commitment to the methodology document. An optional extension lets attestors record their verification or dispute of a measurement, and a further extension records the evolution of methodology over time so that historical measurements retain the method under which they were originally made. Corrections are additive and the correction history cannot fork.

What it contributes

Methodology hash mandatory; latestIndicatorSnapshot() is not chronological latest - use period-aware queries; double-counting prevention is off-chain consumer responsibility; privacy caution for beneficiary/location data on public chains.

Impact data on-chain has prior art. Carbon-credit protocols already track project provenance and retirement, and they work, but they do so for a single asset class with a fixed indicator and a single methodology. This standard contributes the generalisation: a structure that works for any quantitative impact indicator, under any methodology, with the methodology itself versioned and the whole record made tamper evident. The framing is tamper-evidence rather than any claim to prevent misrepresentation outright; the standard cannot stop a false figure being entered, but it can ensure that a figure, once entered, cannot be silently revised, and that any correction is permanently visible to anyone, on any platform.

Methodology accreditors/measurement bodies sit above.

Prior art: Toucan/KlimaDAO and Regen Network work in single-asset-class carbon with fixed methodology. This standard generalises to any quantitative indicator under any methodology - that is the contribution. Note: Does not make SFDR/ISSB deployments compliant by itself.

In use

In institutional investor with a sustainability mandate buys into an asset partly on the strength of its reported environmental performance. A year later the figures have improved, and the investor needs to know whether the asset genuinely performed better or whether the measurement methodology was quietly changed to produce a flattering number. In conventional reporting these two possibilities are hard to separate, which is the space in which greenwashing operates. A methodology-aware, append-only log makes the distinction visible. The investor can see

whether the underlying measurement improved under a constant method, or whether the method itself was superseded, and when.

What is at stake?

Sustainability-linked and impact capital is among the fastest-growing pools of institutional money, and it is also the pool under the closest scrutiny for the credibility of its claims. An impact assertion that cannot be independently verified, and whose methodology can be revised without trace, is a reputational liability rather than an asset to the institution holding it. Unless impact can be made auditable and its methodology made transparent across the market rather than within a single issuer's reporting, this capital cannot deploy into tokenised impact assets with confidence, and the largest natural source of demand for exactly these assets stays on the sidelines.

Subject-Linked NAV Snapshot Oracle

What it does

This standard defines an interface for publishing and reading the net asset value of a subject. Each published value is keyed by currency, carries an explicit basis indicating whether it is expressed per unit, per share, or in total, and records the timestamp at which the valuation itself was struck. A consuming contract can read the value together with two independent staleness signals: one indicating whether a new value has been published within an expected interval, and one indicating whether the valuation itself is recent, regardless of when it was last published. Corrections are additive, and an optional extension supports fixed median aggregation (latestAggregationTimestamp) of values from multiple independent providers with detection of material divergence between them.

What it contributes

Provider = msg.sender; dual staleness (isPublishStale / isValuationStale); terminal-only fixed median aggregation (latestAggregationTimestamp); optional multi-provider divergence detection; methodology commitment may use Standard Two bundle hash for multi-document valuation memos.

Publishing a valuation is not a problem anyone needs help with; fund administrators and data providers do it routinely. The unsolved problem is reading a valuation safely when you did not produce it. A consuming system needs to know not only the number but whether it can be relied upon, and specifically whether it is stale, and in which of two distinct senses, since a provider can republish an old valuation promptly and satisfy a freshness check on publication while the valuation itself is months out of date. This standard contributes the consumer-side safety semantics that existing vault and price-feed standards assume but do not specify: the separation of publication staleness from valuation staleness, the explicit basis of the figure, and the explicit principle that a reported valuation is not a tradeable price. It composes with existing vault standards rather than replacing them, and it makes a valuation safely consumable by any system that had no part in producing it, which is portability of trust applied to the number itself.

Valuation agents/auditors sit above.

This standard composes with ERC-4626 vault interfaces via an adapter or cached-value pattern. Vaults MUST NOT call latestNAVStatus() directly inside convertToAssets() (it can revert). Consumers gate pricing on dual staleness: publication heartbeat and valuation age, computed via safe timestamp subtraction.

In use

A lending protocol wishes to accept a tokenised real-world asset as collateral. To set a safe loan-to-value ratio it must read the asset's current value, and it is exposed to a specific danger - lending against a valuation struck before a material decline in worth, because the number was merely republished rather than refreshed. With an interface that exposes both whether the figure was recently published and whether the underlying valuation was recently struck, the protocol can refuse to price against a stale valuation and protect its lenders.

What is at stake?

A tokenised asset that cannot be valued safely by a system that did not produce the valuation cannot serve as collateral, cannot be priced reliably on a secondary venue, and cannot be integrated into the wider financial machinery that gives an asset its liquidity. The valuation staying readable only within the issuer's own environment is what keeps tokenised real-world assets as isolated holdings rather than instruments that move through a market. The deep, liquid markets that institutional capital requires form only when value can be trusted by parties who had no hand in producing it, which is the precise gap this interface closes.